# Emanuel Azcona, Ph.D.

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## SUMMARY

Lead data scientist and applied ML researcher (Ph.D., Northwestern University) with a proven track record of architecting end-to-end ML and VLM systems that convert unstructured product data into actionable business assets. 5+ years of experience driving technical strategy and development across multimodal (vision-language) AI, graph neural networks, and recommender systems. Scaled data science initiatives adopted by 20+ global teams, across merchandising, product design, and marketplace strategy, facilitating data-driven decisions about future palettes for footwear. Passionate about ethical AI, technical mentorship, and applying ML to benefit consumers.

## INDUSTRY EXPERIENCE

### Lead Data Scientist | *Nike, Inc. – Product Creation & Merchandising Data Science*

(Feb. 2025 – Present) · San Francisco, CA (Remote)

- Productionized a VLM-driven structured extraction pipeline, converting a self-started exploratory DS experiment into a maintained asset that delivers quantitative color insights for **90K+ footwear (FW) products** and continues to be **used by 20+ global teams** to inform merchandising decisions, product briefs, and marketplace strategy with a data-driven lens.
- Reframed color estimation from simple pixel counting to a multimodal, context-aware process, enabling granular, accurate color tagging that complements existing hierarchical product categories.
- Designed the **human-in-the-loop (HITL) evaluation** strategy to assess VLM outputs, combining human-labeled ground truth data, VLM-as-a-judge validation, inter-rater reliability, and **quantitative error metrics rooted in human perception** to establish quality thresholds for production-level data.
- Built scalable VLM inference and evaluation pipelines in Databricks (PySpark/Python) with contract-driven prompts/schemas, MLflow experiment tracking, and quota-aware batching to enable repeatable execution over the full product catalog.
- Translated ambiguous data requirements into robust model constraints/guardrails for high-fidelity DS assets that support cross-functional business partners across several product creation and merchandising workflows.

### Senior Data Scientist | *Nike, Inc. – Product Creation & Merchandising Data Science*

(Feb. 2023 – Feb. 2025) · Chicago, IL (Remote)

- Led a cross-functional build involving DS and ML teams to develop a product similarity engine utilizing a HNSW vector search with metadata-filtered retrieval, producing product substitute tables for downstream planning and assortment teams.
- Adapted an open-source hybrid recommender system to fuse historical engagement signals from digital product channels with perceptual color features (continuous), enhancing personalization relevance beyond discrete product color labels.
- Released and maintained pre-computed product **color metadata consumed by 20+ analytics teams**, enabling color-driven product briefs, merchandising insights, and express lane color palette decision-making **across core marketplaces in FY24.**

### Data Scientist | *Nike, Inc. – Product Creation & Merchandising Data Science*

(Sept. 2022 – Jan. 2023) · Seattle, WA (Remote)

- Built color-theory-grounded image processing pipelines to extract per-product color proportions from display imagery, establishing the foundational metric later adopted across color analytics workstreams.
- Co-developed business logic with merchandising and regional analytics teams to profile footwear products by perceived color blocking, enabling consumer color-preference segmentation.

### Data Scientist & Graduate Date Science Intern | *Nike, Inc. – Sport Activity Data Science*

(June 2021 – Sept. 2022) · Evanston, IL (Remote)

- Developed a graph-based workout recommender system (GNN) for the Nike Training Club app, incorporating biometric and interaction data to surpass parity with the existing production recommender in offline evaluation.
- Surfaced user engagement patterns from user and workout GNN embeddings to inform workout content sequencing, contributing to retention-focused workout improvements and prioritizing achievable and relevant recommendations as positive reinforcement.

### Artificial Intelligence Intern | *Stats Perform*

(June – Aug. 2018) · Chicago, IL

- Optimized data pipelines for accessing and preprocessing multi-agent football tracking data (France Ligue 1), leveraging MongoDB and S3 to streamline ingestion of large-scale spatiotemporal datasets.
- Designed temporal graph convolutional autoencoders to compress sparse player tracking sequences into dense representations, achieving <1 mm. average position reconstruction error and enabling downstream exploration of prediction models for predicting and simulating multi-agent movement in team-based sports.

## ACADEMIC EXPERIENCE

**Adjunct Associate Faculty | *Columbia University***

(Sept. 2025 – Present) · San Francisco, CA (Remote)

- Provided strategic and technical mentorship to student groups in a capstone course (Fall '25), overseeing the end-to-end development of industry-sponsored projects and evaluating the rigor of ML/analytics applications in solving business problems.
- Partnered with faculty lead to deliver instruction, ensuring seamless coordination, learning, and impactful capstone projects.

**Ph.D. Candidate & Research Assistant | *Northwestern University, Image & Video Processing Lab***

(Sept. 2017 – June 2022) · Evanston, IL

- Delivered specialized lectures on 3D imaging and machine learning for as part of a special series of lectures within the Deep Learning Foundations and Image Processing courses, supporting the curriculum for advanced topics in these areas for 100+ students.
- Authored and presented multiple peer-reviewed research publications on advanced graph-based machine learning methods for predicting and characterizing Alzheimer's disease (AD) using multimodal neuroimaging data.
- Developed and applied Graph Neural Network (GNN) frameworks to model cognitive and behavioral phenomena, including depression, addiction, and demographic signals, as networks constructed from psychological picture-rating tasks.
- Designed variational autoencoder (VAE) architectures to learn and visualize latent representations of cortical folding patterns, enabling analysis of structural brain changes across stages of AD and cognitive impairment.
- Built a GNN-based diagnostic classifier achieving 96.35% accuracy in distinguishing AD patients from healthy controls across 300+ subjects, while maintaining interpretability through graph-level and regional analyses.
- Created graph-based 3D brain manifold visualizations to provide region-level explainability and anatomically grounded insights into model predictions for neurological disease.
- Managed shared high performance GPU infrastructure for a 12+ researcher lab, optimizing resource allocation, scheduling, and utilization across multiple concurrent ML research workflows.

## EDUCATION

— **Ph.D. Electrical Engineering, 2022,** *Northwestern University*
Dissertation: *Geometric Deep Learning in Neuroimaging and Human Reward Behavior*

— **M.S. Electrical Engineering, 2019,** *Northwestern University*

— **B.S. Electrical Engineering, 2017, *cum laude,*** *New York University*
Senior Thesis: *Predicting NBA Playoff Contention Through Individual Player Performance*

## TECHNICAL SKILLS

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| **Languages:** | Python, SQL, C++, Matlab, Bash, LaTeX, HTML/CSS |
| **DS & ML:** | PyTorch, Scikit-learn, Pandas, Tensorflow/Keras, PyTorch Geometric (PyG), Matplotlib/Seaborn, Docker |
| **Data & Cloud:** | PySpark, AWS (S3, Sagemaker Studio), Databricks, MongoDB, Google Colab |
| **Domains:** | Machine learning, medical imaging , graph neural networks, recommender systems, embedding analysis, computer vision, image processing, convolutional neural networks, transformers, vision language models (VLMs), prompt engineering |

## COMMUNICATION & LEADERSHIP

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| **Languages:** | Full professional proficiency in English and Spanish (bilingual). |
| **Documentation Excellence:** | Commended by peers for delivering high-impact, cohesive technical documentation that bridges the gap between machine learning (ML) research and scalable industry application. |
| **Workshop/Demo Facilitation:** | Architect and deliver technical workshops on multimodal AI (VLM) and GenAI, prioritizing clarity for both technical and non-technical audiences. |
| **Academic Lecturing:** | Experienced in translating ML topics into digestible course material at varying levels ranging from high school to graduate and professional-level classrooms, focusing on the rigorous understanding and communication of fundamentals. |